

KSP Crime Intelligence Platform (V8)

AI-Driven Crime Analytics & Visualization Platform

Team: My internet is Cooked

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Abstract

The KSP Crime Intelligence Platform is a next-generation predictive policing dashboard built for the Karnataka State Police. Unlike generic heatmaps, this platform does not simply predict where crime *might* happen. It uses strict mathematical modeling to tell station commanders exactly how much to **trust** a prediction, optimizes patrol deployment for maximum spatial efficiency, and actively guards against systemic community bias.

1 Project Overview

Live Deployment

- **Dashboard:** [Zoho Catalyst AppSail Deployment](#)
- **Health Check API:** `/api/health`

2 Core Features & Technical Implementation

The backend intelligence is driven by a 9-phase analytical pipeline (`ksp_pipeline_v8_final.py`), running over a synthetic dataset of 80,000 records across 164 stations in 31 districts.

2.1 Phase 1 & 2: Spatial Hotspots & Baselines

- **DBSCAN Clustering:** Uses the Haversine metric to cluster recent incidents geographically, identifying immediate, unpredicted flare-ups.
- **Seasonal Baseline:** Computes a naive seasonal benchmark (4-week rolling mean + 52-week lag) to serve as a strict evaluation floor for the machine learning models.

2.2 Phase 3: Two-Headed Forecasting Engine

We utilize `HistGradientBoosting` trees from `scikit-learn` to power two distinct heads:

1. **Count Regressor:** Trained with a Poisson loss function (`loss="poisson"`) to predict the exact number of incidents expected next week.
2. **Severity Classifier:** A binary classifier calibrated using Platt Scaling (`LogisticRegression`) to output the strict mathematical probability ($P(\textit{Serious})$) that at least one crime in the predicted cluster will be a serious offence (e.g., dacoity, murder).

2.3 Phase 4: Near-Repeat Space-Time Contagion (Knox Test)

Crime behaves like a contagion. If a house is burglarized, neighboring houses are at statistically higher risk for the next 72 hours.

- We utilize the **Knox space-time permutation test**.
- Risk is calculated using a dual exponential decay kernel:

Formula

$$W = \exp\left(-\frac{\Delta t}{\tau}\right) \exp\left(-\frac{d}{\sigma}\right) \quad (1)$$

Where τ is time-decay and σ is spatial-decay. This auto-flags stations at risk of “contagious” crime sprees.

2.4 Phase 5: Statistical Smoothing (Empirical Bayes)

Small-number noise in rural stations causes severe false alarms (e.g., an increase from 1 to 2 crimes looks like a 100% surge). We fix this using **Gamma-Poisson Empirical Bayes Smoothing**.

Formula

The historical rate is smoothed toward the district mean using a posterior update:

$$\alpha_{\text{post}} = \alpha_0 + \sum y_i \quad ; \quad \beta_{\text{post}} = \beta_0 + n \quad (2)$$

This generates an **80% Bayesian Credible Interval**. If the interval is wide, the station is rendered on the map with a *dashed border*, warning the commander that confidence is low.

2.5 Phase 6: Prospective PAI Calibration

Standard machine learning “accuracy” is a misleading metric for crime prediction. Instead, we optimize for the **Predictive Accuracy Index (PAI)**.

Formula

$$\text{PAI} = \frac{n/N}{a/A} \quad (3)$$

(Where n = crime in flagged zones, N = total crime, a = flagged area, A = total area).

- Using **Voronoi tessellation** to approximate station patrol areas, the pipeline dynamically calibrates unique risk thresholds for all 31 districts.

Result

Our untouched test-period median PAI is **1.319**, proving the model is 31.9% more spatially efficient than random patrol deployment.

2.6 Phase 7: Descriptive Fairness Audit

Predictive policing often creates feedback loops. Phase 7 executes a strict audit across 9 demographic/community groups (SC, ST, OBC categories, General, Minority).

Result

The pipeline enforces a tight **1.40 percentage point spread** between the lowest (34.94%) and highest (36.34%) arrest rates across all 9 groups, yielding a Min/Max ratio of 0.96 (passing the strict > 0.80 four-fifths rule).

2.7 Phase 8 & 9: Cyber Network & Anomaly Detection

- **Cyber Routes:** Tracks repeat offenders (`AccusedMasterID`) across 3,600+ synthetic cyber FIRs using a bipartite network graph, connecting the dots on organized syndicates hitting multiple jurisdictions.
- **FDR Anomaly Detection:** Uses Poisson upper-tail testing with Benjamini-Hochberg False Discovery Rate control ($\text{FDR } \alpha = 0.05$) to alert on statistically significant weekly anomalies.

3 User Interface (The Dashboard)

The frontend (`index.html`) is a single-page, responsive web application offering three primary intelligence views:

- **Map + Simulator:** An interactive state map plotting all 164 stations (Red = HOTSPOT, Orange = WATCH, Green = SAFE). It includes a **Scenario Simulator** allowing commanders to inject holiday spikes or adjust time-of-day intensity to see dynamic predictive changes without retraining the AI.
- **Cyber Routes:** A node-and-link network graph filtering cyber FIRs to track repeat syndicates. Clicking an accused node draws an inferred, station-level route map of their activity.
- **Audit & Trends:** Exposes the underlying health of the AI. Displays the fairness metrics, PAI test scores, and 9 critical pipeline sanity checks.

4 Technology Stack

- **Frontend:** HTML5, Vanilla CSS, JavaScript (`Chart.js`, `Leaflet.js`, `Cytoscape.js`)
- **Backend Pipeline:** Python 3.10+
- **Machine Learning & Math:** `scikit-learn` (`HistGradientBoosting`), `scipy` (`Voronoi`/`Spatial`/`Stats`), `numpy`, `pandas`
- **Geospatial Processing:** `shapely`, `haversine`
- **Hosting & Deployment:** Zoho Catalyst (AppSail Serverless Hosting)

5 Repository Structure

```
ksp-crime-intelligence/
|-- MasterCodeBase/                # Core V8 Pipeline & AppSail Server
|   |-- ksp_pipeline_v8_final.py    # The core 9-phase ML pipeline script
|   |-- server.py                  # AppSail Python HTTP Server for the dashboard
|   |-- index.html                 # The frontend dashboard UI
|   |-- pipeline_outputs/          # Generated metrics, models, and JSON payloads
|   |-- reports/                   # Final findings, sanity checks, and benchmark re
|-- data/                          # Project Datasets
|   |-- Dataset_grounded.csv        # 80,000-record synthetic incident dataset
|   |-- ksp_station_registry_v8.csv # 164-station geographical registry
|   |-- dashboard_data_v8.json      # Compiled JSON payload used by the frontend
|-- presentation/                  # Pitch deck and Datathon slide assets
|-- backend/                       # Scaffolding for future FastAPI / Postgres integ
|-- frontend/                      # Scaffolding for future React integration
|-- docs/                          # Project documentation
'-- README.md
```

6 Execution & Deployment

6.1 Running the ML Pipeline Locally

To regenerate the models, metrics, and JSON data for the dashboard:

```
cd MasterCodeBase
pip install -r requirements.txt
python ksp_pipeline_v8_final.py
```

This script acts as a reproducible artifact. It executes all 9 phases, runs the 9 sanity checks, and outputs `dashboard_data.json` directly into the `data/` folder for the frontend to consume.

6.2 Secure AppSail Refresh

When deployed to Catalyst, the pipeline can be triggered to refresh the data securely via the `/api/pipeline/run` endpoint. This requires `ENABLE_PIPELINES_RUN=1` and a secure `X-Admin-Token` header. Rebuilds are **atomic**, ensuring zero downtime for officers using the dashboard.

7 Ethical Guardrails & Project Limitations

- **No “Black Box” Trust:** Stations with high predictive uncertainty are rendered with a *dashed border* on the map, signaling to commanders that the prediction is a lead, not a certainty.
- **No Device Tracking:** Cyber routes are built strictly from historical FIR station-filing records. It explicitly is *not* device, IP, or real-time location surveillance.
- **Descriptive, Not Punitive:** The fairness module reports observed demographic variances. It deliberately avoids slapping a simplified “bias score” on the dashboard without actual population denominators, promoting transparent oversight.

- **Synthetic Data Disclaimer:** This repository uses synthetic data. It is an engineering demonstration of advanced spatial modeling, not a claim about real Karnataka crime patterns.